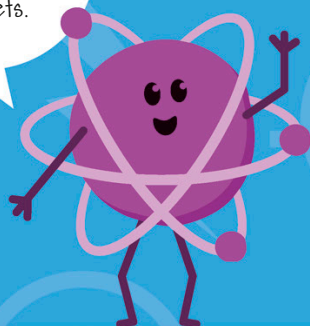


The electricity that comes to your home is dangerously powerful. **Never** put your fingers in electrical sockets.

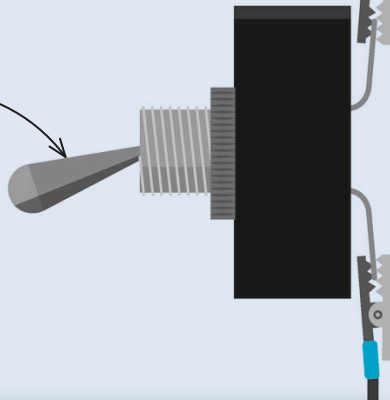


Wires

The current flows through a loop made of metal wires.

Switch

Putting a switch in a circuit lets you control the current. With a switch, you can turn the flow of electricity on or off.



Switch control

A switch is something that can break the loop of an electric circuit. Turning the switch off opens a gap in the loop. Turning the switch on closes the gap.

Switch off

The loop is broken. Electricity cannot travel across the gap in the circuit, so the bulb will not light up.



Switch on

The loop is complete. With the gap closed, electricity flows around the circuit and the bulb shines.



Conductors

A material that electricity can flow through easily is called a **conductor**. Metals are good conductors. Water also conducts electricity well, so make sure you do not touch electrical objects with wet hands—stay safe!



Copper wire

Copper is a type of metal, and it is a very good conductor. Copper wire is often used to connect electrical components in a circuit.



Graphite

Pencil "lead" is made of a material called graphite. Crumbly graphite is not a metal, but it does conduct electricity.

Insulators

Materials that block the flow of electric current are known as **insulators**. Good insulators include rubber, wood, wool, glass, air, and plastic. Insulators are used to keep electrical equipment safe.



Rubber

Electricians often wear rubber gloves and boots when working. Rubber does not conduct electricity, so it stops them from getting electric shocks.



Plastic

Plastics are used to coat wires and cables to stop electricity from leaking out. They are also used to make electrical plugs and sockets.